

L^AT_EX Counters and Lengths

1 Defining and Setting New Counters

1.1 Basic Counters Operations

The Initial Value is 0

Setting a Value for the Counter Now the value becomes 15

Stepping the Counter Now the value becomes 16

Adding a Constant Value Now the value becomes 20

Decreasing the Counter Value Now the value becomes 19

Doubling the Counter Value Now the value becomes 38

1.2 Correlated/Hierarchical Counters

Using the `\newcounter{follower}[leader]`

Value of the *No* Counter defaults to 0 also.

Setting the New Counter *No* value to be 5, and keep in mind that the leader is of value 38

Adding to the Follower to be 10

Stepping the Follower to be 11

Changing the Leader to Reset the Follower Now the leader is 39, and follower is 0

Desired Typeset of the Follower 39.0

Using the `\counterwithin{follower}{leader}`

Set the Leader The Leader Value is 5

Print the Follower The Initial print of the Follower is 5.0, Zero of course.

If stepped The Follower will be 5.1

Now, stepping the Leader to reset the Follower to 6.0 zero again.

The Leader and The Follower are 6, and 6.0

Here the value of *Within* is 1.2.5, Linked to the parent subsection.

1.3 New Subsection Leader to Reset Within

Here the value of *Within* changes to 1.3.0 to match the subsection number. We're still in the coupled mode, so *Within* is 1.3.8

The *Within* is now 8

2 Methods of Printing Counters Value

1. Arabic 7
2. Lower Case roman vii
3. Upper Case Roman VII
4. Lower Case Letter g
5. Upper Case Letter G
6. Foot Note Style **

2.1 Using the *fntcount* Package

Ordinal 24th

Number String TWENTY-FOUR

Ordinal String twenty-fourth and Twenty-Fourth and TWENTY-FOURTH

2.2 The `\value` cmd

The value of one is 1

Now value of *one* is 2

This command is used to access the value of certain counter in order to use inside a latex command for some purpose like adding to a counter. It is useful for doing arithmetic with counters. you use `\value{count}` for situations where LaTeX is expecting to process a numeric value. It is intended for use within other commands to access the value of the counter as a TeX number.

3 Practical Examples of Using Counters

3.1 New Structure Containing Counter

Theorem I. *Gravity or gravitation, is a natural phenomenon by which all things with mass or energy—including planets, stars, galaxies, and even light are brought toward (or gravitate toward) one another. On Earth, gravity gives weight to physical objects, and the Moon’s gravity causes the ocean tides.*

Theorem II. Optional Headline *The gravitational attraction of the original gaseous matter present in the Universe caused it to begin coalescing and forming stars and caused the stars to group together into galaxies, so gravity is responsible for many of the large-scale structures in the Universe. Gravity has an infinite range, although its effects become weaker as objects get further away.*

3.2 Creating a New Ordered List

I Optional First Item.

II Second Item.

III Third Item.

IV Fourth Item.

V Fifth Item.

I First Item Again.

II Second Item Again.

3.3 Different Approach for New Lists

A First Entry.

B Head Second Entry.

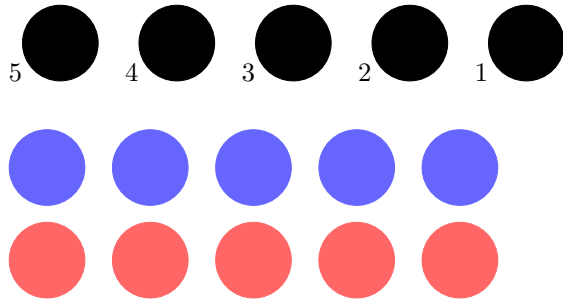
C Third Entry.

Optional Title for the List

A First Ball Entry.

B **Descriptive Head** Second Ball Entry.

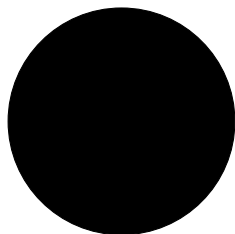
4 Looping: While Statement



5 L^AT_EX Lengths

5.1 Creating, Setting, and Using New Length

My *Span* is 50.00mm , which is 5.000 cm



5.2 Setting Length Dimensions to Some Value

Some Width

In that sense, you have the with of [Some Width](#) equaling to 1.923 cm.

Similarly, you have `\settoheight`
 2^{4^6} , and this height is ≈ 10 pt

5.3 Using Lengths for Custom Structures

Some Illustration Example

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason.
